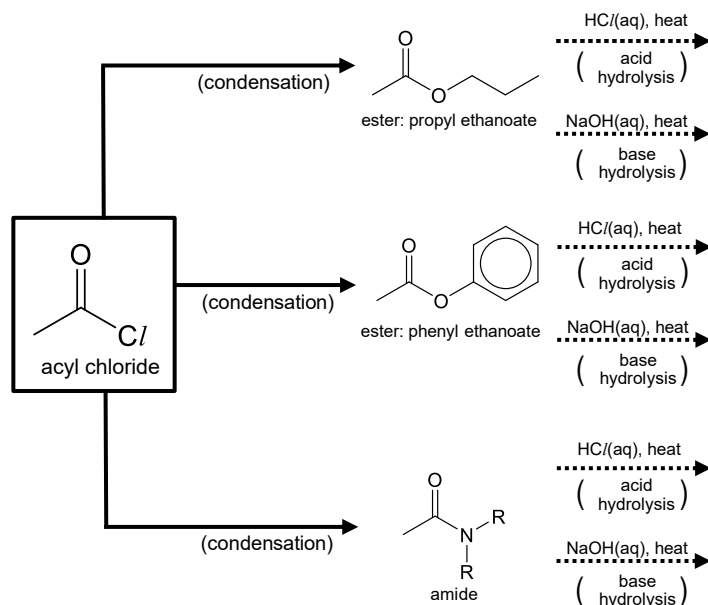
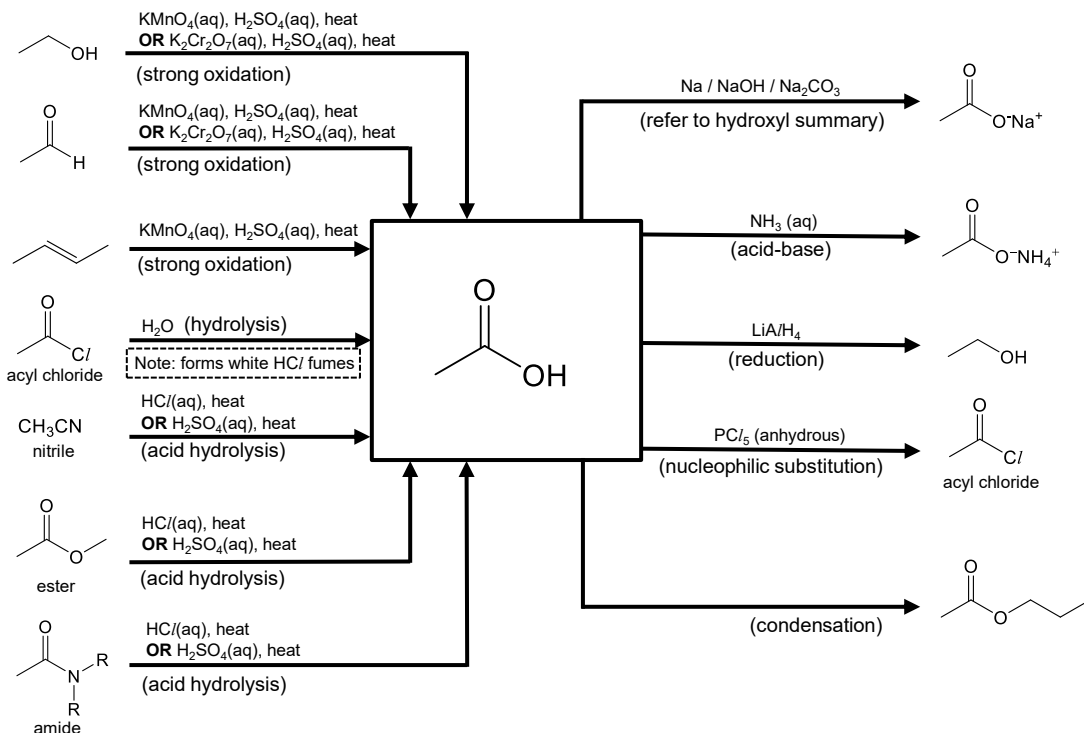


Reactions



Relative acidity

Rank the following from least to most acidic

<chem>c1ccccc1O</chem>	<chem>CCC(=O)O</chem>	<chem>ClCCC(=O)O</chem>	<chem>CC(C)(Cl)C(=O)O</chem>

Explanation:

➤ The more _____ the conjugate base, the _____ the acid.

- | | |
|---|--|
| <p><chem>c1ccccc1[O-]</chem></p> <ul style="list-style-type: none"> p orbital of O⁻ overlaps with π e⁻ cloud of benzene ring lone pair of e⁻ on O⁻ | <p><chem>ClCCC(=O)[O-]</chem></p> <ul style="list-style-type: none"> Cl atom is _____ further disperses negative charge on O⁻, hence even more stabilised |
| <p><chem>CCC(=O)[O-]</chem></p> <ul style="list-style-type: none"> disperse negative charge on O⁻, hence conjugate base stabilised p orbital of O⁻ overlaps with π e⁻ cloud of C=O bond negative charge _____ disperse negative charge on O⁻ to a greater extent, hence more stabilised | <p><chem>CC(C)(Cl)C(=O)[O-]</chem></p> <ul style="list-style-type: none"> _____ of Cl atom to -COO⁻ group further disperses negative charge on O⁻, hence most stabilised |

Ease of hydrolysis

Rank the following in order of increasing ease of hydrolysis

<chem>c1ccccc1Cl</chem>	<chem>CC(C)(Cl)C</chem> assume S _N 2	<chem>CC(=O)Cl</chem>

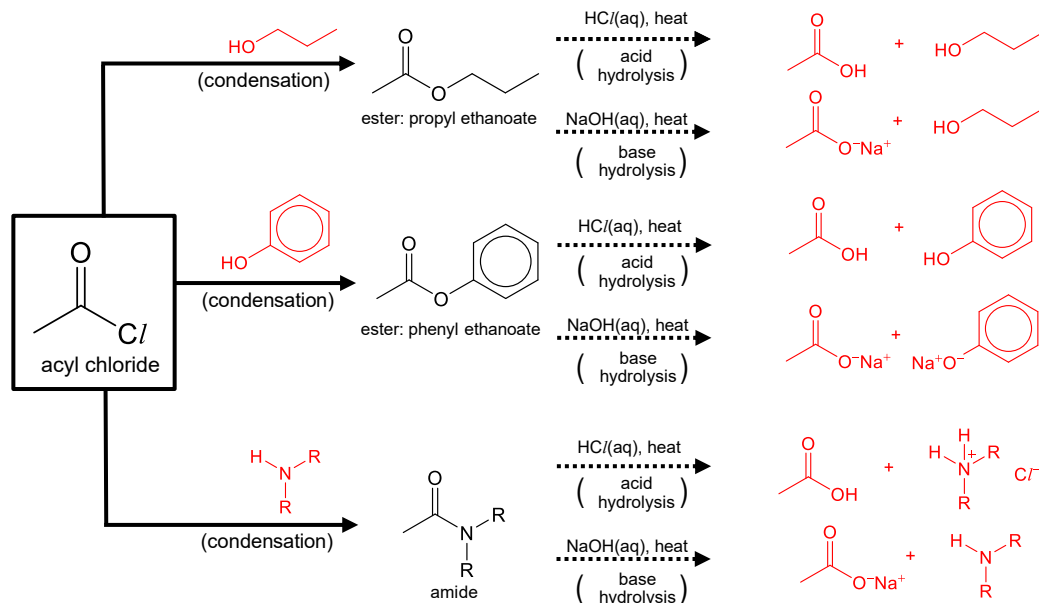
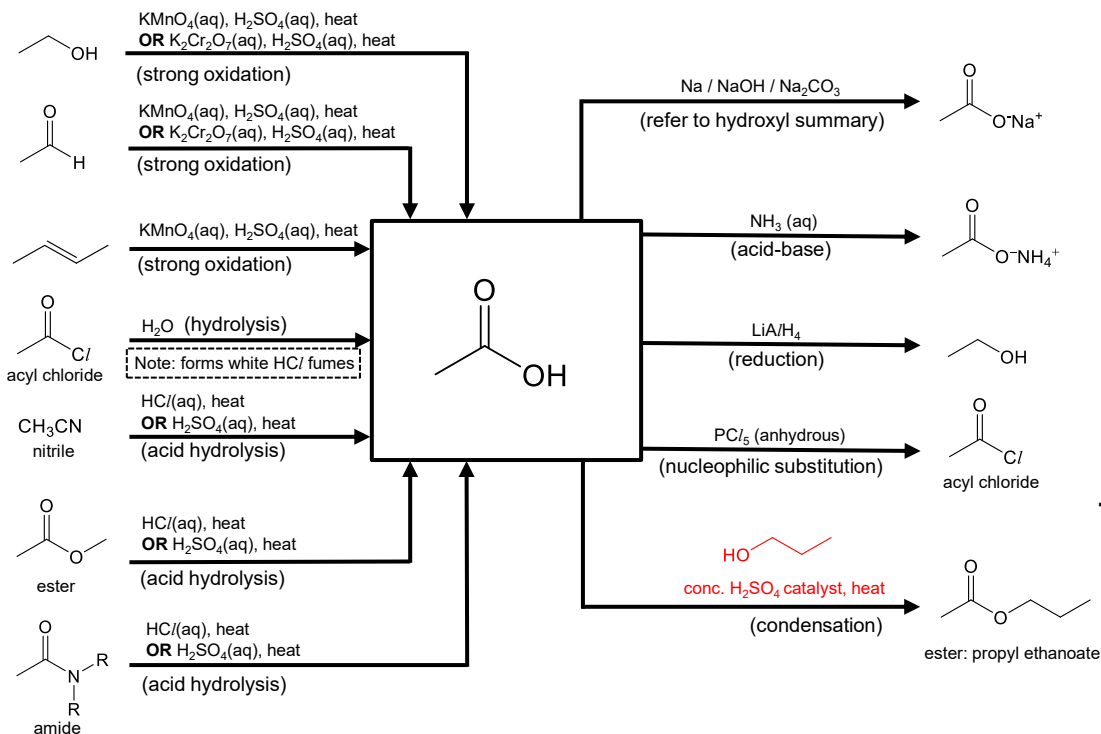
Explanation:

① **Electronic factor:** The more electron-_____ () the C, the more susceptible it is to the attack by H₂O nucleophile.

② **Steric factor:** The _____ steric hindrance about the δ⁺ C, the more susceptible it is to the attack by H₂O nucleophile.

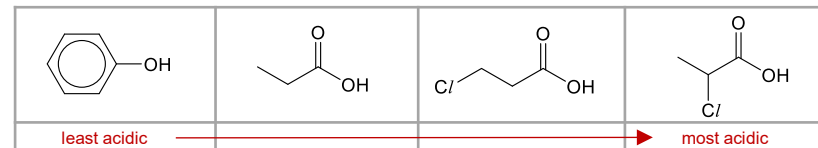
- | | | |
|--|---|--|
| <p><chem>c1ccccc1Cl</chem></p> <ul style="list-style-type: none"> _____ C-Cl bond (partial double bond character) π e⁻ cloud of benzene ring repels nucleophile | <p><chem>CC(C)(Cl)C</chem></p> <ul style="list-style-type: none"> C is slightly δ⁺ because bonded to only one electronegative Cl atom _____ C, more steric hindrance about δ⁺ C | <p><chem>CC(=O)Cl</chem></p> <ul style="list-style-type: none"> C is very δ⁺ because bonded to two electronegative atoms; O and Cl _____ C, less steric hindrance about δ⁺ C |
|--|---|--|

Reactions



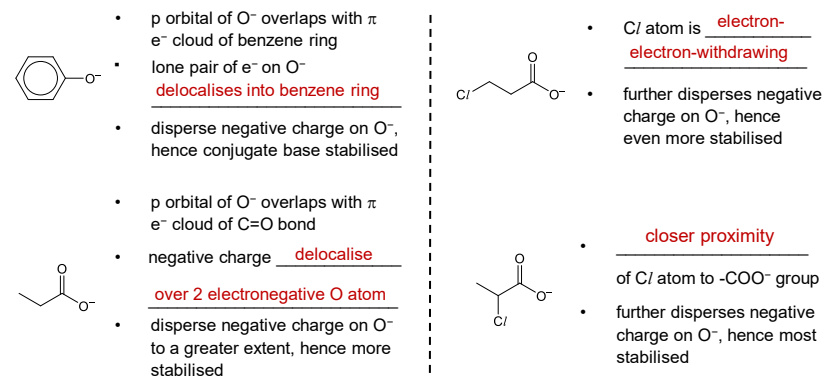
Relative acidity

Rank the following from least to most acidic



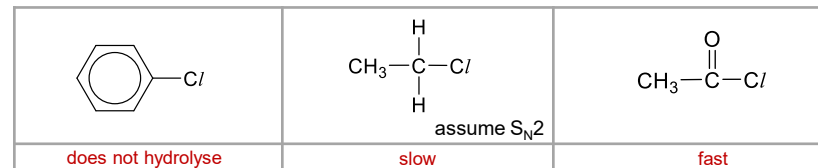
Explanation:

➤ The more stable the conjugate base, the stronger the acid.



Ease of hydrolysis

Rank the following in order of increasing ease of hydrolysis



Explanation:

① **Electronic factor:** The more electron-deficient (δ^+) the C, the more susceptible it is to the attack by H_2O nucleophile.

② **Steric factor:** The less steric hindrance about the δ^+ C, the more susceptible it is to the attack by H_2O nucleophile.

